

Mohammed Afaan Ansari

College Park, MD | (240) 413-1682 | mansari1@umd.edu | linkedin.com/in/afaan001 | github.com/affan00733

Data Scientist | ML Modeling, Anomaly Detection & Applied Research. Designed anomaly-detection & sustainability metrics used by FAA, MSCI, and Mayo Clinic; cut latency 40% and serving cost 99.98%. ASONAM 2023 *2nd Best Paper*; lead author COLM 2026.

EDUCATION

University of Maryland, College Park

M.S. Computer Science | GPA: 3.81/4.0 | Advisor: Prof. C. Buntain (CLIP/iSchool)

Aug 2024 – May 2026

College Park, MD

Sardar Patel Institute of Technology, Mumbai University

B.Tech. Computer Engineering | GPA: 8.9/10 | ACM/Springer publications: NLP & blockchain

Jul 2019 – Jun 2022

Mumbai, India

TECHNICAL SKILLS

- **ML & Statistics:** Python, PyTorch, scikit-learn, TensorFlow, pandas, NumPy; anomaly detection, classification, regression, NLP, time-series, CV; A/B testing, hypothesis testing, causal inference, feature engineering.
- **Data & Cloud:** SQL (BigQuery, Postgres), PySpark; Matplotlib, Seaborn, Tableau; GCP (Vertex AI, BigQuery), AWS (SageMaker, S3), Docker, Airflow, REST APIs, CI/CD; ONNX-quantized model deployment.

EXPERIENCE & RESEARCH

CSSI Inc.

Data Science Intern

Jun 2025 – Aug 2025

Washington, D.C.

- Designed anomaly-detection pipeline on 10M+ FAA aviation-weather records (90%+ recall), enabling analysts to detect hazardous flight patterns 2× faster; built FAA-compliant ETL + APIs.

MSCI Sustainability Institute

Research Participant — Green CapEx Initiative

Jul 2025 – Sep 2025

Remote

- Computed ESG/CapEx metrics from 10K+ records adopted into MSCIs Green CapEx taxonomy used by institutional investors.

University of Maryland — CLIP Lab

Graduate Research Assistant (Prof. C. Buntain)

Aug 2024 – Present

College Park, MD

- Shipped multimodal summarizer in RAG pipeline (10K+ docs); cut analyst turnaround 12× (2h→15m) and doubled crisis-report throughput for 20+ analysts.

MSCI Inc.

Data Science Technology Associate | Star Performer 2023, top of 1,200+

Jul 2022 – Aug 2024

Mumbai, India

- Promoted to Associate in 1yr (vs. typical 3–4); scaled GCP pipeline on 100K+ research reports (QA errors -30%); cut serving cost \$10→\$0.002/1K (**99.98% reduction**).

Georgia Tech & Mayo Clinic

Research Assistant (Prof. Joy Arulraj)

Dec 2022 – Dec 2023

Atlanta, GA (Remote)

- Developed MRI tumor-detection model (85% accuracy on 1K patients) with Mayo Clinic; benchmarked GPT-4 vs. ONNX inference in EVADB (150ms→90ms).

Rochester Institute of Technology

Research Assistant — Social Data Mining (Prof. A. KhudaBukhsh)

Jul 2022 – Jun 2023

Rochester, NY (Remote)

- Co-authored ACM/IEEE paper on tweet lifespan using BERT; *2nd Best Paper, ASONAM 2023*; cited in subsequent social-data studies.

Sourceable Inc. (Columbia-backed startup)

Lead Software Developer

Jan 2022 – Jul 2024

New York, NY (Remote)

- Directed crisis-mapping platform (React, Docker, K8s) adopted by 1,500+ journalists across 52+ countries.

Nomura Holdings — Global Markets IT

Technology Analyst Intern

Jan 2022 – Jun 2022

Mumbai, India

- Engineered ELK-based monitoring system for trading infrastructure; secured full-time return offer with 40% salary uplift.

SELECTED PROJECT & LEADERSHIP

- **ChartDesign (COLM 2026, lead author)** — LoRA-tuned LLMs (CSV → chart JSON); **84% accuracy**; 92% LLM-human agreement; 65% preferred over human charts.
- Winner — Univ. Hackathon (LLM QnA), Pharma Hackathon; mentored 40+ students (10+ at FAANG); raised \$50K for NGO.